AI + Quantum AI Senior Engineer Learning Plan

Senior-Level Full-Stack AI + Quantum AI — Learning Plan

Owner: MD Hashim · **Window:** 12 May 2026 → 31 Dec 2026 · **Budget:** ~30 hrs/week (≈990 hrs total)

How to read this plan

You already ship in production across drug discovery ML, VQE/ADAPT-VQE/DMET-VQE, RAG, and biomedical segmentation. The gap you described is **theory + mathematical intuition + the *why* behind techniques** — exactly what gets probed in senior interviews and project-design discussions. This plan is built around that gap, not the syntax.

Three things to internalize before starting:

1. **Math runs *parallel* to everything else.** It's not a separate phase you "finish." Roughly 6–8 hrs/week of every week is math (Track M below). The remaining 22–24 hrs goes to the phase topic.
2. **Watch → Re-derive on paper → Code from scratch → Read the paper.** Video is your on-ramp, not your destination. Every concept gets re-derived on paper (this is the senior-level habit you need to build) and re-implemented in a stripped-down form before you move on.
3. **Interview & whiteboard checkpoints** are sprinkled in. Treat them as forcing functions. If you can't whiteboard the derivation cold, you don't yet own the concept.

Phases overlap intentionally — the parallel math track and the immediate-need segmentation work both start in Week 1.

Track M — Mathematical Foundations (Parallel, ALL phases)

~7 hrs/week throughout, ~230 hrs total

This is your weakness-by-self-report, so we front-load it. Heavier in weeks 1–10, lighter (but never zero) afterward.

M1. Linear Algebra (Weeks 1–4, heavy; then reference)

The single highest-leverage topic for both deep learning AND quantum.

- **3Blue1Brown — Essence of Linear Algebra** (15 videos, ~3 hrs total — watch this in Week 1, twice if needed) https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab
- **MIT 18.06 — Linear Algebra, Gilbert Strang** (the gold standard; pick lectures, don't binge all 35) https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/video_galleries/video-lectures/ Priority lectures for you: 1, 3, 5, 6, 9, 10, 14, 15, 16, 21, 22, 25, 29, 30
- **MIT 18.065 — Matrix Methods in Data Analysis, Strang** (this is the *applied* version — SVD, PCA, low-rank, gradient descent through a linear algebra lens — closer to ML reality) https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/video_galleries/video-lectures/

Whiteboard checkpoint (end Week 4): Derive SVD from eigendecomposition. Explain *geometrically* what a rank-r approximation does to a matrix. Show why `A^T A` is always PSD. Derive backprop through a linear layer using matrix calculus.

M2. Multivariable Calculus + Matrix Calculus (Weeks 3–6)

- **3Blue1Brown — Essence of Calculus** https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr
- **The Matrix Cookbook** (reference PDF — keep open while reading any paper) https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf
- **Matrix Calculus for Deep Learning** (Parr & Howard — best short article on Jacobians for ML) https://explained.ai/matrix-calculus/

M3. Probability + Information Theory (Weeks 5–10)

- **Harvard Stat 110 — Joe Blitzstein** (~34 lectures; the best intro-prob course period; skip if you're already comfortable through MGFs and conditional expectations) https://projects.iq.harvard.edu/stat110/youtube
- **MIT 6.041 — Probabilistic Systems Analysis, John Tsitsiklis** (alternative if Blitzstein feels slow) https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/
- **Information Theory — David MacKay book chapters 1–6** (entropy, KL, mutual info — required for understanding loss functions, VAEs, diffusion, contrastive learning) https://www.inference.org.uk/itila/book.html (free PDF)

Whiteboard checkpoint (end Week 10): Derive cross-entropy from KL divergence. Explain why minimizing NLL = MLE. Derive the ELBO. Explain mutual information and where it shows up in InfoNCE / contrastive losses.

M4. Optimization (Weeks 7–14)

- **Stanford EE364A — Convex Optimization, Stephen Boyd** (the canonical course; even non-convex DL builds on these intuitions) https://www.youtube.com/playlist?list=PL3940DD956CDF0622 Companion textbook (free PDF): https://web.stanford.edu/~boyd/cvxbook/
- **Sebastian Ruder — Gradient Descent Optimization Algorithms** (single best summary article) https://www.ruder.io/optimizing-gradient-descent/

M5. Group Theory & Symmetry (Weeks 12–18, lighter; required for quantum + geometric DL)

You need just enough group theory to read VQE/quantum chemistry papers AND geometric deep learning papers without panic.

- **Group Theory for Physicists — Brian Hill (YouTube)**, or **Eddie Munoz "Group Theory" playlist** — pick one, ~10–15 hrs https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures
- **Geometric Deep Learning — Bronstein et al. (course videos)** (Lie groups, equivariance — pays off in Phase 5) https://geometricdeeplearning.com/lectures/

PHASE 1 — Medical Image Segmentation Mastery

Weeks 1–6 (12 May → 22 Jun 2026) · ~22 hrs/wk on this phase + 8 hrs Track M · ~135 phase hours **Goal:** Own MRI / fluorescence brain segmentation end-to-end — physics → preprocessing → architectures → losses → 3D → uncertainty → evaluation. By end of phase you should be able to whiteboard nnU-Net's design choices from memory and defend each one.

Week 1 — Imaging physics + MRI fundamentals (you cannot do brain MRI without this)

Most engineers skip this and it shows in their model design. You won't.

- **Allen Elster — Questions & Answers in MRI** (text reference, browse 2–3 hrs) https://mriquestions.com/index.html
- **OHBM Educational — "MRI Physics" lecture playlist** (search YouTube: "OHBM MRI physics for beginners") https://www.youtube.com/@ohbmonline/playlists
- **Paul Callaghan — "Magnetic Resonance" Otago lectures** (deeper physics — pick first 4 lectures) https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z
- **Fluorescence microscopy primer** — iBiology playlist (your tissue work needs this) https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMl3PlpdW9TMP

Deliverable: A 1-page personal note explaining T1 vs T2 vs FLAIR vs T1CE, what brain tissue looks like in each, why each modality exists, and what artifacts (motion, bias field, partial volume) you must preprocess. For fluorescence: explain photobleaching, channel bleed-through, autofluorescence, and how they affect segmentation.

Week 2 — Classical segmentation, U-Net family, and the math of convolutions

- **Stanford CS231n 2017 — Justin Johnson** (full course, but focus this week on Lectures 5, 9, 11) Playlist: https://www.youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLfQRF3EO8sYv Course notes (excellent): https://cs231n.github.io/
- **AI for Medical Diagnosis — Pranav Rajpurkar (deeplearning.ai / Coursera)** — *structured medical-imaging warmup; the brain MRI segmentation lab in Week 3 of the course is directly aligned with your work. Audit free or use the trial. Skip the X-ray module if pressed for time; focus on segmentation + evaluation.* https://www.coursera.org/learn/ai-for-medical-diagnosis

- **Original U-Net paper (Ronneberger 2015)** — read line-by-line, derive output sizes by hand https://arxiv.org/abs/1505.04597
- **V-Net (3D), Attention U-Net, U-Net++, ResUNet, nnU-Net** — read in this order, 2 hrs each:
  - V-Net: https://arxiv.org/abs/1606.04797
  - Attention U-Net: https://arxiv.org/abs/1804.03999
  - nnU-Net (the *most important* paper for you, read twice + supplementary): https://www.nature.com/articles/s41592-020-01008-z

Whiteboard checkpoint: Draw U-Net from memory. Compute output size at every layer for a 256×256 input. Explain *why* skip connections work (gradient flow + spatial recovery). Explain *why* nnU-Net's "no new architecture" thesis is actually profound.

Week 3 — Loss functions for medical segmentation (your 95/5 class-imbalance problem)

This is where most segmentation pipelines die. You need to own this cold.

- **A survey of loss functions for semantic segmentation (Jadon)** — read & derive each loss https://arxiv.org/abs/2006.14822
- **Boundary Loss (Kervadec et al.)** — for thin structures / boundary precision https://arxiv.org/abs/1812.07032
- **Tversky / Focal Tversky** — your friend for severe imbalance https://arxiv.org/abs/1810.07842
- **Generalized Dice + compound losses** — Sudre 2017: https://arxiv.org/abs/1707.03237

Coding deliverable: From-scratch PyTorch implementations of: Dice, GDL, Focal, Tversky, Focal Tversky, Boundary, and a compound `DiceCE` loss. Unit-test on synthetic imbalanced data. Plot gradients vs. class ratio. Document when each loss helps (this is interview gold).

Week 4 — nnU-Net deep dive + MONAI ecosystem

By now nnU-Net stops being a black box.

- **MONAI tutorials GitHub** (clone, run, modify all the brain segmentation notebooks) https://github.com/Project-MONAI/tutorials
- **DigitalSreeni (Sreenivas Bhattiprolu) — Python for Microscopists channel** — *the single most practical, microscopy-native channel for your work; far more aligned with fluorescence + volumetric biomedical imaging than any Stanford course* Channel: https://www.youtube.com/@DigitalSreeni/playlists Code repo (matches every video): https://github.com/bnsreenu/python_for_microscopists Priority playlists for your stack: U-Net for image segmentation (https://www.youtube.com/playlist?list=PLZsOBAyNTZwbR08R959iCvYT3qzhxvGOE), the 3D U-Net for volumetric segmentation videos (search his channel for "3D U-Net"), and his fluorescence microscopy / multi-channel image videos. Pick ~6–8 videos that map to your current bottlenecks; do them code-along, not passive.
- **MONAI YouTube channel** (~3 hrs of focused content) https://www.youtube.com/@projectmonai/videos
- **nnU-Net v2 official docs + code walkthrough** (read the source — it teaches more than the paper) https://github.com/MIC-DKFZ/nnUNet
- **MONAI + nnU-Net integration** (nnUNetV2Runner) https://github.com/Project-MONAI/tutorials/blob/main/nnunet/README.md
- **Kitware's MONAI Auto3DSeg vs nnU-Net comparison** https://www.kitware.com/developing-custom-3d-medical-image-segmentation-solutions-using-out-of-the-box-pipelines-in-monai/

Deliverable: Reproduce nnU-Net on a public brain dataset (BraTS subset or any task from Medical Segmentation Decathlon). Write up: what defaults did it pick, *why* those defaults, what would you change for fluorescence microscopy where statistics differ from MRI?

Week 5 — Transformer-based segmentation + Foundation models (SAM, MedSAM)

This is where the field is now and where senior interview questions are heading.

- **Vision Transformer paper (ViT)** — re-derive attention math https://arxiv.org/abs/2010.11929
- **TransUNet** — https://arxiv.org/abs/2102.04306
- **Swin-UNet** — https://arxiv.org/abs/2105.05537
- **UNETR / Swin UNETR** (3D, MONAI implements both) — https://arxiv.org/abs/2103.10504, https://arxiv.org/abs/2201.01266
- **SAM (Segment Anything)** — https://arxiv.org/abs/2304.02643
- **MedSAM** — https://www.nature.com/articles/s41467-024-44824-z

- **MedNeXt (ConvNet that beats transformers on medical seg in many cases)** — https://arxiv.org/abs/2303.09975

Whiteboard checkpoint: Compare inductive biases of CNNs vs ViTs vs hybrid. When does each win? What's the role of pretraining/foundation models in low-data medical regimes? Walk through SAM's promptable design and what makes MedSAM different.

Week 6 — 3D segmentation, uncertainty, evaluation, and clinical realities

3D convolution intuition (close this gap before you touch 3D U-Net seriously):

- **DigitalSreeni — "3D U-Net for semantic segmentation"** (architecture walkthrough + tensor shapes) https://www.youtube.com/playlist?list=PLZsOBAyNTZwbR08R959iCvYT3qzhxvGOE (search his channel for the 3D U-Net videos in particular)
- **CS231n lecture on 3D CNNs / video CNNs** (in the 2017 playlist, the "videos & 3D" lecture)
- **Convolution arithmetic paper (Dumoulin & Visin)** — the canonical reference for shapes/strides/padding/dilation in 2D *and* 3D https://arxiv.org/abs/1603.07285 Companion animations: https://github.com/vdumoulin/conv_arithmetic
- **Whiteboard exercise:** for a 3D conv with input (D=64, H=128, W=128, C_in=4), kernel (3,3,3), stride (1,2,2), padding (1,1,1), output channels 32 — compute the output tensor shape AND the parameter count by hand. Repeat for transposed conv and dilated conv.
- **3D medical imaging — TorchIO library + tutorials** (data augmentation done right for 3D) https://torchio.readthedocs.io/ Video: https://www.youtube.com/playlist?list=PLfXIfDgnRPlflEHFx2N6mlbn-MGFOPiX2
- **Uncertainty in deep learning (Yarin Gal's thesis + talks)** https://www.youtube.com/results?search_query=yarin+gal+uncertainty+deep+learning Practical: MC Dropout, Deep Ensembles, evidential learning
- **Metrics Reloaded** (the definitive paper on choosing eval metrics — published 2024, must-read) https://www.nature.com/articles/s41592-023-02151-z
- **Domain shift in medical imaging** — Stanford AIMI talks https://aimi.stanford.edu/education/educational-resources

Phase 1 capstone deliverable (end Week 6): A self-contained Jupyter notebook + 3-page write-up on YOUR mouse brain dataset (or a public BraTS subset if your data isn't transportable). The write-up has 4 sections: **(1)** preprocessing decisions with physics justification, **(2)** architecture choice + ablation across 3 architectures, **(3)** loss function ablation, **(4)** uncertainty quantification & failure-mode analysis. This is the kind of artifact a senior engineer produces; it's also exactly what an interview panel wants to see.

Phase 1 interview self-test (do this before moving on):

- Whiteboard the U-Net forward pass + skip connections including tensor shapes.
- Derive the Dice gradient and explain why Dice is unstable for empty masks.
- Explain why nnU-Net beats fancier architectures most of the time (and when it doesn't).
- Explain attention mathematically and walk through ViT's patch embedding.
- For a new dataset, walk through how you'd diagnose: bias field artifacts, class imbalance, label noise, domain shift.

PHASE 2 — Modern Deep Learning Theory + Foundation Architectures

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Weeks 7–12 (23 Jun → 3 Aug 2026) · ~23 hrs/wk on phase + 7 hrs Track M · ~140 phase hours **Goal:** Stop using transformers as a black box. By end of phase you implement GPT-from-scratch, derive every layer of attention, understand modern training tricks (RoPE, FlashAttention, mixture of experts, RMSNorm), and can defend architecture choices on theoretical grounds.

Weeks 7–9 — Andrej Karpathy's "Neural Networks: Zero to Hero" (THE most important DL series for senior depth)

This is non-negotiable. Even with your experience, doing all 10 videos *and* the exercises will rebuild your foundations from first principles.

- **Playlist:** https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ
- **Course page:** https://karpathy.ai/zero-to-hero.html

- Schedule (do them in order, with exercises):
  - Week 7: micrograd → makemore part 1 → makemore part 2 (MLP)
  - Week 8: makemore parts 3, 4, 5 (BatchNorm internals, manual backprop, WaveNet)
  - Week 9: GPT from scratch, tokenization (tiktokenizer), Let's reproduce GPT-2
- After each video: re-implement WITHOUT looking at his code. This part is what makes the difference.

Week 10 — Transformers in depth + modern variants

- **Stanford CS25 — Transformers United (current edition)** — pick 6–8 talks https://web.stanford.edu/class/cs25/ YouTube playlist: https://www.youtube.com/playlist?list=PLoROMvodv4rNiJRchCzutFw5ItR_Z27CM
- **Lilian Weng's blog** (every senior ML eng reads this; pick attention, transformer, diffusion, RLHF posts) https://lilianweng.github.io/
- **The Annotated Transformer** (Sasha Rush — code + math line by line) http://nlp.seas.harvard.edu/annotated-transformer/
- **Modern training tricks** (each is a 30-min paper read):
  - RoPE: https://arxiv.org/abs/2104.09864
  - RMSNorm: https://arxiv.org/abs/1910.07467
  - FlashAttention: https://arxiv.org/abs/2205.14135
  - GQA / MQA: https://arxiv.org/abs/2305.13245
  - SwiGLU: https://arxiv.org/abs/2002.05202
  - Mixture of Experts (Switch Transformer): https://arxiv.org/abs/2101.03961

Whiteboard checkpoint: Derive scaled dot-product attention from scratch. Explain *why* the $\sqrt{d_k}$ scaling. Derive multi-head as a single matmul. Explain RoPE geometrically. Explain why pre-LN is more stable than post-LN.

Week 11 — Vision Transformers, hybrid models, and ConvNets in the modern era

- **Lucas Beyer's ViT lectures + threads** (he co-authored ViT) https://www.youtube.com/results?search_query=lucas+beyer+vision+transformer
- **ConvNeXt** (CNNs vs ViTs, surprisingly nuanced): https://arxiv.org/abs/2201.03545
- **DINOv2** (self-supervised ViT — relevant to your medical work where labels are scarce) https://arxiv.org/abs/2304.07193
- **MAE (Masked Autoencoder)**: https://arxiv.org/abs/2111.06377
- **Hugging Face Computer Vision course** (good as a structured companion) https://huggingface.co/learn/computer-vision-course

Week 12 — Diffusion models + generative modeling theory

Even if you're not doing GenAI right now, diffusion math underpins the modern field and shows up in medical image synthesis/augmentation.

- **Stanford CS236 — Deep Generative Models, Stefano Ermon** (rigorous; watch lectures 1–3, 6–8, 11–13) https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8
- **Jonathan Ho et al. — DDPM (original)**: https://arxiv.org/abs/2006.11239
- **Yang Song — Score-based generative models (the unification)** + his blog post https://yang-song.net/blog/2021/score/
- **Lilian Weng — "What are Diffusion Models?"** (the consolidating read) https://lilianweng.github.io/posts/2021-07-11-diffusion-models/
- **Hugging Face Diffusion Course** (do unit 1 & 2 hands-on) https://huggingface.co/learn/diffusion-course

Phase 2 interview self-test:

- Implement attention in 5 minutes on a whiteboard.
- Derive the loss for a diffusion model (variational + score-matching views).
- Explain why we use LayerNorm not BatchNorm in transformers (this is asked all the time).
- Discuss positional encodings: sinusoidal vs learned vs RoPE vs ALiBi — when each helps.
- Compute the parameter count of a transformer block given d_model, n_heads, FFN ratio.

PHASE 3 — Quantum Computing + Quantum Machine Learning (PROJECT-CRITICAL)

Weeks 13–22 (4 Aug → 12 Oct 2026) · ~22 hrs/wk phase + 8 hrs Track M · ~220 phase hours **Goal:** Match your existing Lilly hands-on work (VQE/ADAPT-VQE/DMET-VQE) with rigorous theory. By end of phase you should be able to derive VQE from the variational principle, explain barren plateaus mathematically, defend ansatz choices, and discuss QML's current limits honestly (per Maria Schuld's recent critiques).

Important: your resume shows you've already shipped VQE/ADAPT-VQE/DMET-VQE work. This phase is **theory & depth**, not from-zero. You should be re-deriving things you already used, and reading the critical-perspective literature so you can speak to the field honestly in interviews.

Week 13 — Quantum mechanics math (Hilbert spaces, operators, tensor products)

- **MIT 8.04 — Quantum Physics I, Allan Adams** (skip ahead to lectures 6–12 if you find the experimental motivation slow) https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/video_galleries/lecture-videos/
- **Frederic Schuller — Geometric Anatomy of Theoretical Physics** (for the rigorous-math types — optional but transforms your view) https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic
- **Quantum Country** (the spaced-repetition intro — text-based, surprisingly fast and sticky) https://quantum.country/qcvc

Whiteboard checkpoint: Define a Hilbert space rigorously. Tensor-product two qubit states. Explain unitarity, Hermitian operators, and why measurement outcomes are eigenvalues. Distinguish pure vs mixed states; explain the density matrix.

Weeks 14–16 — Quantum computing fundamentals (the IBM/Watrous track)

The IBM Quantum Learning courses by John Watrous are the best free quantum-info series available. Do all four.

- **Basics of Quantum Information** — John Watrous (IBM Quantum Learning) https://learning.quantum.ibm.com/course/basics-of-quantum-information YouTube: https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2IqmSaUjfnhvBHR
- **Fundamentals of Quantum Algorithms** — Watrous https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms
- **General Formulation of Quantum Information** — Watrous (density matrices, channels, POVM — important for noise/error mitigation) https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information
- **Foundations of Quantum Error Correction** — Watrous https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction
- **Companion textbook (free):** Nielsen & Chuang — *Quantum Computation and Quantum Information* (chapters 1–4, 8, 10)

Weeks 17–18 — Variational quantum algorithms: VQE, ADAPT-VQE, QAOA (your bread and butter, now with theory)

You've shipped this — now own the math.

- **PennyLane Codebook — Variational Quantum Algorithms** https://pennylane.ai/codebook
- **PennyLane VQE tutorials** (work through code; you've done this in practice — now read all derivations) https://pennylane.ai/qml/demos_quantum-chemistry
- **Original VQE paper — Peruzzo et al. 2014** — re-derive https://www.nature.com/articles/ncomms5213
- **ADAPT-VQE — Grimsley et al. 2019** — re-read carefully https://www.nature.com/articles/s41467-019-10988-2
- **DMET-VQE / fragment methods** (your stack) — survey paper: https://arxiv.org/abs/2108.08987
- **QAOA — Farhi 2014** + Hadfield's review https://arxiv.org/abs/1411.4028 https://arxiv.org/abs/1709.03489

Whiteboard checkpoint: Derive the variational principle. Walk through VQE end-to-end: ansatz choice (hardware-efficient vs UCCSD vs ADAPT), gradient estimation (parameter shift rule), classical optimizer choice. Explain barren plateaus mathematically and the techniques to mitigate them. Why is ADAPT-VQE more parameter-efficient? When is DMET fragmentation worth the overhead?

Weeks 19–20 — Quantum Machine Learning (variational circuits, kernels, quantum neural networks)

- **PennyLane "Quantum Machine Learning" topic page (Maria Schuld curated)** — the most honest overview in the field https://pennylane.ai/topics/quantum-machine-learning
- **Maria Schuld — "Taking stock of QML, a critical perspective"** (this is the talk every senior should watch — it tells you what's hype and what's real) https://www.youtube.com/results?search_query=maria+schuld+taking+stock+quantum+machine+learning
- **Schuld & Petruccione — *Machine Learning with Quantum Computers* (Springer 2021)** — chapters 4–7 (kernels, variational, QNNs)
- **Christophe Péré's QML course (École de Technologie Supérieure)** — full lectures + notebooks https://github.com/Christophe-pere/QML-Course
- **Morten Hjorth-Jensen — Quantum Computing & ML course (University of Oslo)** — recent 2026 semester recordings on YouTube https://github.com/CompPhysics/QuantumComputingMachineLearning
- **Key papers to read (one per evening):**
    - "Quantum machine learning models are kernel methods" (Schuld 2021): https://arxiv.org/abs/2101.11020
    - Barren plateaus (McClean 2018): https://www.nature.com/articles/s41467-018-07090-4
    - Expressibility & entangling capability (Sim 2019): https://arxiv.org/abs/1905.10876
    - "Is quantum advantage the right goal for QML?" (Schuld & Killoran 2022): https://arxiv.org/abs/2203.01340
    - PQC encodings: https://arxiv.org/abs/2008.08605

Weeks 21–22 — Quantum chemistry on quantum computers (your domain) + Qiskit Global Summer School recordings

- **Qiskit Global Summer School 2025 — full recordings** (the recent edition's quantum chemistry & QDA tracks are critical for your work) https://www.youtube.com/@qiskit/playlists
- **IBM Quantum chemistry course materials on IBM Quantum Learning** https://learning.quantum.ibm.com/
- **Sample-based Krylov / Quantum Diagonalization Algorithms (QDA)** — newest practical near-term technique IBM has open-sourced https://www.ibm.com/quantum/blog/iql-migration
- **Pauli decomposition + measurement grouping** (key for VQE shot-budget engineering): https://arxiv.org/abs/1907.13117

Phase 3 capstone deliverable (end Week 22): A self-contained notebook + 4-page write-up: pick *one* molecule from your existing Lilly work (e.g. the keto-enol tautomerism case or HATU coupling). Re-implement it three ways — vanilla VQE with UCCSD, ADAPT-VQE, and DMET-VQE. Benchmark against classical CCSD/6-31G. The write-up must defend EVERY choice on theoretical grounds: ansatz family, optimizer, measurement grouping, noise mitigation, basis set. Include a barren-plateau analysis. This document is exactly what a senior QML interview wants to see.

Phase 3 interview self-test:

- Derive VQE from the variational principle.
- Explain the parameter-shift rule and why it works.
- Whiteboard a barren plateau argument.
- When would you use QAOA vs VQE? When would you NOT use a quantum algorithm at all?
- Honest answer: where is QML overhyped and where does it have real promise (you should be able to cite Schuld here).
- Walk through DMET fragmentation and the classical-quantum interface.

PHASE 4 — Full-Stack AI / Production Systems

Weeks 23–28 (13 Oct → 23 Nov 2026) · ~24 hrs/wk phase + 6 hrs Track M · ~145 phase hours **Goal:** Senior-level production skill across LLM systems, RAG, MLOps, distributed training, serving, and ML systems design. You've shipped these; this phase upgrades from "I built it" to "I can defend every architectural choice and design the next-gen version."

Week 23 — LLM internals (post-Karpathy continuation)

- **Karpathy — "Let's reproduce GPT-2" + "Let's build the GPT Tokenizer"** (in the same Zero-to-Hero playlist)
- **Hugging Face NLP course** (units 1–7 — pick what's new to you) https://huggingface.co/learn/nlp-course
- **Stanford CS336 — Language Modeling from Scratch (Percy Liang, Tatsu Hashimoto — 2024+ edition)** — currently the best LLM-systems course https://stanford-cs336.github.io/spring2024/ YouTube: https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_

Week 24 — RAG done at senior level (you've shipped this — now go deeper)

- **Hamel Husain — "What we've learned from a year of building with LLMs"** (the field's most cited senior-eng essay; Part I and Part II) https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/
- **Eugene Yan — patterns for building LLM systems** https://eugeneyan.com/writing/llm-patterns/
- **Pinecone / Anthropic / Voyage — contextual retrieval techniques** Anthropic contextual retrieval: https://www.anthropic.com/news/contextual-retrieval
- **LlamaIndex / LangChain advanced retrieval tutorials** https://docs.llamaindex.ai/
- **Reranking, hybrid search, query rewriting, evals** — go deep on each
- **RAGAS — eval framework** (essential for production RAG) https://docs.ragas.io/
- **DSPy** (programmatic prompting — the modern alternative to manual prompt engineering) https://dspy.ai/

Week 25 — MLOps + ML systems design

- **Chip Huyen — *Designing Machine Learning Systems*** (book — but also her talks are excellent) https://huyenchip.com/ml-interviews-book/contents/8.1.1-ml-systems-design.html
- **Stanford CS329S — Machine Learning Systems Design** https://stanford-cs329s.github.io/ YouTube: https://www.youtube.com/results?search_query=stanford+cs329s
- **Made With ML — MLOps course** (free, hands-on) https://madewithml.com/
- **Full Stack Deep Learning** (free archived course; excellent) https://fullstackdeeplearning.com/course/2022/

Week 26 — Distributed training, serving, inference optimization

- **Sebastian Raschka — Distributed training & efficient fine-tuning** (the best practical writeups) https://magazine.sebastianraschka.com/
- **Hugging Face — Efficient training on multiple GPUs** https://huggingface.co/docs/transformers/en/perf_train_gpu_many
- **vLLM / TGI / SGLang — inference servers** (read the papers + run them) vLLM paper: https://arxiv.org/abs/2309.06180
- **Quantization deep dive — GPTQ, AWQ, GGUF, FP8** — pick papers + Tim Dettmers's writeups
- **DeepSpeed ZeRO + FSDP** — read both papers ZeRO: https://arxiv.org/abs/1910.02054

Week 27 — Evaluation, observability, safety, alignment basics

- **Anthropic's Responsible Scaling Policy + Constitutional AI papers** https://www.anthropic.com/news/constitutional-ai
- **Eugene Yan — LLM Evals** https://eugeneyan.com/writing/llm-evaluators/
- **LangSmith / Phoenix / Helicone — observability tools** (just pick one, run it on a real RAG system)

Week 28 — ML systems design interview prep

Senior interviews increasingly include system design rounds. Practice these end-to-end.

- **Chip Huyen's ML interviews book** (free online) https://huyenchip.com/ml-interviews-book/
- **Practice prompts:** "Design a RAG-based assistant for HCPs that handles 10K queries/day on 50K documents." "Design a brain-segmentation pipeline that ingests new institution data weekly with domain shift handling." "Design retraining + monitoring for a drug-discovery digital twin."
- Practice these on whiteboard, recorded, then critique yourself.

PHASE 5 — Specialization (Geometric DL / Drug-Discovery AI) + Integration + Interview Prep

Weeks 29–33 (24 Nov → 31 Dec 2026) · ~24 hrs/wk phase + 6 hrs Track M · ~120 phase hours **Goal:** Plug the geometric-deep-learning and molecular-AI gap that connects your imaging, quantum, and drug-discovery work. Then integrate everything and run mock interviews.

Week 29 — Geometric Deep Learning foundations

- **Michael Bronstein et al. — Geometric Deep Learning course (AMMI 2022 / 2023)** — lectures available free https://geometricdeeplearning.com/lectures/
- **The "GDL proto-book"** (Bronstein, Bruna, Cohen, Velickovic) — free PDF https://arxiv.org/abs/2104.13478
- **Petar Veličković — GNN lectures** https://www.youtube.com/results?search_query=petar+velickovic+gnn+lectures

Week 30 — Molecular ML + cheminformatics ML

- **Stanford CS279 — Computational Biology** (relevant lectures)
- **DeepChem tutorials** (you may know basics — go deeper) https://deepchem.io/tutorials/
- **Therapeutics Data Commons (TDC)** — benchmark datasets + tutorials https://tdcommons.ai/
- **Key papers:**
  - SchNet (continuous-filter conv for molecules): https://arxiv.org/abs/1706.08566
  - DimeNet++ (directional message passing): https://arxiv.org/abs/2011.14115
  - MolFormer / chemical foundation models: https://arxiv.org/abs/2106.09553
  - Equivariant networks (Cohen, Welling): https://arxiv.org/abs/1602.07576

Week 31 — Protein structure (AlphaFold) + equivariant networks

This connects everything: it's geometric DL, foundation models, biology, and quantum chemistry in one place.

- **DeepMind — AlphaFold 2 explainer lecture (Mohammed AlQuraishi)** — outstanding https://www.youtube.com/results?search_query=AlphaFold+2+explained+AlQuraishi
- **AlphaFold 2 paper**: https://www.nature.com/articles/s41586-021-03819-2
- **ESM / ESMFold (Meta's protein LLM)** — papers + repo
- **e3nn / EGNN — equivariant networks** (papers + tutorials)
- **AlphaFold database (you have access via MCP skill in your tools)** — practical exploration of predicted structures

Weeks 32–33 — Integration, mock interviews, portfolio polish

This phase is where you become *interview-ready as a senior engineer*, not just technically loaded.

- **Build one integration project** that touches your three domains (e.g., "Predict molecular property using a hybrid quantum-classical model with classical pretraining on a chemical foundation model; deploy as a service with proper monitoring"). This is your portfolio centerpiece.
- **Mock interviews (record yourself):**
  - 60-min ML system design — give yourself a prompt, whiteboard it on camera, watch back, critique. Repeat 5 times across different prompts.
  - 45-min technical deep-dive on segmentation (your current project). Imagine an interviewer asking "why this loss, why this architecture, what would you change."
  - 45-min on quantum (defend your VQE / ADAPT-VQE / DMET design choices to a skeptical theoretical chemist).
  - 30-min "tell me about a project" structured behavioral — STAR format, prep 4 stories.
- **Read 1 paper per day** for these two weeks from your subfields. Force the habit before the new year.
- **Write 2 LinkedIn / personal-blog posts** distilling something you've learned. The act of writing for an audience exposes shaky understanding. Don't skip this.

Weekly time allocation (default template — adjust per phase)

Activity	Hours/week
Phase video lectures + paper reading	14
Hands-on coding / re-implementation	7
Math foundation (Track M)	7
Whiteboard re-derivation + note-taking	2
Total	**30**

Cross-cutting habits (do these every week, no exceptions)

1. **Paper-of-the-week.** One paper, deeply read, derivations re-done on paper. Don't break the streak.
2. **Whiteboard Friday.** Pick one concept from the week and whiteboard it without notes. Photograph and keep — these become your interview prep deck.
3. **Re-implementation Saturday.** One concept implemented from scratch in <200 lines. Repo: `weekly-rebuild/wk01-unet`, etc.
4. **Sunday writeup.** 1-page note in your own words explaining the week's deepest concept *to a hypothetical junior*. Teaching is the senior-level forcing function.
5. **Update your senior-engineer "war chest"** — a single Notion / Obsidian doc collecting: math derivations, paper one-liners, gotchas, anti-patterns. By December you'll have 100+ pages and you'll never panic in an interview again.

Resources index (quick lookup)

Math

- 3B1B Linear Algebra: https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab
- 3B1B Calculus: https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr
- MIT 18.06 Strang: https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/
- MIT 18.065 Matrix Methods: https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/
- Stat 110 (Harvard): https://projects.iq.harvard.edu/stat110/youtube
- Boyd Convex Optimization: https://www.youtube.com/playlist?list=PL3940DD956CDF0622 · https://web.stanford.edu/~boyd/cvxbook/
- MacKay Information Theory: https://www.inference.org.uk/itila/book.html
- Matrix Cookbook: https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf
- Matrix Calculus for DL: https://explained.ai/matrix-calculus/
- Geometric DL: https://geometricdeeplearning.com/lectures/

Medical imaging

- MRI Q&A reference: https://mriquestions.com/index.html
- OHBM educational: https://www.youtube.com/@ohbmonline/playlists
- iBiology microscopy: https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMl3PlpdW9TMP
- AI for Medical Diagnosis (Rajpurkar, deeplearning.ai): https://www.coursera.org/learn/ai-for-medical-diagnosis
- DigitalSreeni (Python for Microscopists) channel: https://www.youtube.com/@DigitalSreeni/playlists · Code: https://github.com/bnsreenu/python_for_microscopists
- Convolution arithmetic paper: https://arxiv.org/abs/1603.07285 · Animations: https://github.com/vdumoulin/conv_arithmetic
- MONAI tutorials: https://github.com/Project-MONAI/tutorials
- MONAI YouTube: https://www.youtube.com/@projectmonai/videos
- nnU-Net v2: https://github.com/MIC-DKFZ/nnUNet
- TorchIO: https://torchio.readthedocs.io/
- Stanford AIMI: https://aimi.stanford.edu/education/educational-resources
- CS231n 2017: https://www.youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLfQRF3EO8sYv

Deep Learning core

- Karpathy Zero to Hero: https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ
- Karpathy hub: https://karpathy.ai/zero-to-hero.html
- Stanford CS25 Transformers: https://web.stanford.edu/class/cs25/
- Stanford CS336 Language Modeling: https://stanford-cs336.github.io/spring2024/
- Stanford CS236 Generative: https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8
- Lilian Weng blog: https://lilianweng.github.io/
- Annotated Transformer: http://nlp.seas.harvard.edu/annotated-transformer/
- HF NLP: https://huggingface.co/learn/nlp-course

- HF CV: https://huggingface.co/learn/computer-vision-course
- HF Diffusion: https://huggingface.co/learn/diffusion-course

Quantum

- IBM Quantum Learning: https://learning.quantum.ibm.com/
- Watrous YouTube series: https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2IqmSaUjfnhvBHR
- Qiskit YT channel: https://www.youtube.com/@qiskit
- PennyLane Codebook: https://pennylane.ai/codebook
- PennyLane QML topic: https://pennylane.ai/topics/quantum-machine-learning
- PennyLane chemistry demos: https://pennylane.ai/qml/demos_quantum-chemistry
- MIT 8.04 QM: https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/
- Quantum Country: https://quantum.country/qcvc
- Hjorth-Jensen QC+ML course: https://github.com/CompPhysics/QuantumComputingMachineLearning
- Christophe Péré QML: https://github.com/Christophe-pere/QML-Course

Production / MLOps

- CS329S systems design: https://stanford-cs329s.github.io/
- Made With ML: https://madewithml.com/
- Full Stack DL: https://fullstackdeeplearning.com/course/2022/
- Chip Huyen ML Interviews: https://huyenchip.com/ml-interviews-book/
- Hamel Husain LLM essay: https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/
- Eugene Yan patterns: https://eugeneyan.com/writing/llm-patterns/
- Anthropic contextual retrieval: https://www.anthropic.com/news/contextual-retrieval
- RAGAS: https://docs.ragas.io/
- DSPy: https://dspy.ai/

Drug discovery / Geometric DL

- Bronstein GDL: https://geometricdeeplearning.com/lectures/
- GDL proto-book: https://arxiv.org/abs/2104.13478
- DeepChem tutorials: https://deepchem.io/tutorials/
- TDC: https://tdcommons.ai/

A note on the discomfort you described

You said it's been "very tough and overwhelming to manage things in core engineering problems with very noble use cases where I need to have in-depth understanding of underlying math and their intuitions." That feeling is correct, and shared by basically everyone making the senior-engineer transition in deep tech. You've shipped a *lot* by pattern-matching and excellent execution. The next level requires you to *derive* what you used to *use* — and that derivation work is slow, frustrating, and feels like going backwards for the first 6–8 weeks. It isn't. Stay on the math track. By Week 10 you'll feel it click, and by Week 20 you'll wonder how you ever read papers without the foundations. Trust the process; the plan is conservative on hours specifically so you don't burn out.

Good luck. Adjust this plan as you go — it's a scaffold, not a prison.